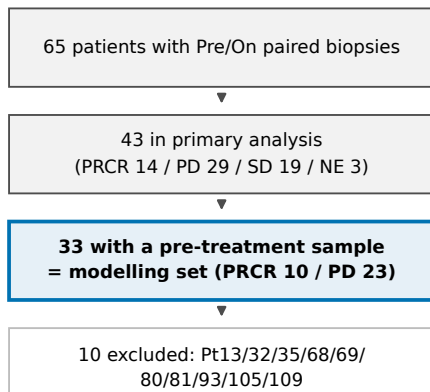


(a) Cohorts and their roles

cohort	smp / pat	role in this study
GSE91061	109 / 65	development
GSE78220	28 / 26	external validation
GSE215868	105 / 105	large-scale validation
GSE244982	41 / 41	direction consistency
GSE308433-435	42 / 34	paired tumour-TCR

(b) Analysis unit: patient



(c) Analysis-unit checks performed

situation	handling
Repeated biopsies	paired / mixed structures; never independent samples
Pt27A / Pt27B	one patient, two anatomical sites; collapsed by median
Pt16 (GSE78220)	on-treatment only; dropped by include_primary flag
Mel-TIL-026 / HM026	the same patient in two cohorts; de-duplicated first
GSE215868 plate ID	plate position, not a patient ID

(d) Why cohorts are never pooled

1. Sampling time GSE91061 pairs pre-dose with C1D29; the others are pre-tx
2. Label definition verbatim RECIST strings vs. best overall response vs. BOR
3. Platform, units raw counts / FPKM / featureCounts / NanoString counts

Consequence: cross-cohort work is limited to direction-consistency tests and external validation, so a cohort difference can never be mistaken for a biological signal.